

SSD_WATCHDOG

usage reference

1 Overview

Watchdog uses a standard linux framework to provide hardware watchdog. The upper-layer application can set the time out time and keep alive by itself.

Watchdog is turned off by default, and you can decide whether to turn it on or not. It is recommended to operate in the main thread when it is turned on. If it is operated in other threads, the watchdog will be closed as the thread is closed.

2. Watchdog Instructions for Use

2.1. Instructions for use under Kernel

2.1.1. Open Watchdog

Open the /dev/watchdog device and the watchdog will be started.

The reference code is as follows:

```
int wdt_fd = -1;
wdt_fd = open("/dev/watchdog", O_WRONLY);
if (wdt_fd == -1)
```

```
{
    // fail to open watchdog device
}
```

2.1.2. Close Watchdog

The reference code is as follows:

```
int option = WDIOS_DISABLECARD ;

ioctl ( wdt_fd ,  WDIOC_SETOPTIONS ,  &option );

if ( wdt_fd  !=  - 1 )
{
    close ( wdt_fd );
    wdt_fd  =  - 1 ;
}
```

2.1.3. Set timeout

The timeout is set by the standard IOCTL command WDIOC_SETTIMEOUT. The unit is second. The timeout time is recommended to be greater than 5s. The reference code is as follows:

```
#define WATCHDOG_IOCTL_BASE 'W'

#define WDIOC_SETTIMEOUT _IOWR(WATCHDOG_IOCTL_BASE, 6, int)

int timeout = 20;

ioctl(wdt_fd, WDIOC_SETTIMEOUT, &timeout);
```

2.1.4. KEEP ALIVE

通过标准的IOCTL命令WDIOC_KEEPAIVE来喂狗，喂狗时间按照设定的timeout来决定，喂狗时间应该比timeout小，参考代码如下：

```
#define WATCHDOG_IOCTL_BASE 'W'

#define WDIOC_KEEPAIVE _IOR(WATCHDOG_IOCTL_BASE, 5, int)

ioctl(wdt_fd, WDIOC_KEEPAIVE, 0);
```

2.2. Instructions for use under Uboot

Include header files: `alkaid\boot\drivers\mstar\wdt\wdt.h` call the following interface

```
void wdt_reset(void);  
void wdt_stop(void);  
void wdt_start(U32 u32LaunchTim);  
void wdt_set_timeout(U32 u32Timeout);
```